Auburn Roses

Lectern One

A Governance Research Programme
on the Hallucination Tax

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Part I

The Hypothesis

The Hallucination Tax

When an AI system produces incorrect output, the human operator must spend time identifying the error, correcting it, and verifying that the correction did not introduce new errors. This cost is the hallucination tax.

For elementary tasks, the tax is low. A misspelled word is caught immediately. A wrong date is checked in seconds. The error is local, the correction is obvious, and the verification is trivial.

For frontier mathematics, the tax is catastrophic. A single undetected error in a proof can invalidate every downstream conclusion. Detecting the error requires expertise that may exceed the operator's own. Correcting it may require restructuring the entire argument. Verifying the correction may require re-auditing everything that depended on the original claim. The cost is not proportional to the error — it is proportional to the complexity of the material the error is embedded in.

The hallucination tax scales with difficulty. And for sufficiently difficult material, the tax can exceed the value of the AI's contribution entirely.

This is the central problem of the AI era. Not capability. Governance.

Why This Problem

If AI is to be the next growth engine — for science, medicine, for industry, for the decisions that shape whether civilisations endure — then the outputs it produces must be trustworthy for their stated purpose. Not plausible. Not impressive. Trustworthy.

The question that launched this programme was not: *can I solve a Millennium Prize Problem?* The question was:

> If we take one of humanity's hardest open questions and systematically reduce the hallucination tax across the landmark available systems — what do they produce? Is the result convergent? Is it verifiable? Is it something that experts can use? Or is it sophisticated sophistry?

The 3D incompressible Navier–Stokes existence and smoothness problem was chosen because it is maximally difficult. Not because the goal was to resolve it. Because a methodology stress-tested on easy problems proves nothing. A methodology that produces stable, verifiable, cross-model convergence on a problem this hard — that is evidence the methodology works. And if convergence can be achieved, the days of hallucination are numbered.

The Governance Thesis

This corpus is not presented as a resolution of Navier–Stokes. It is a governance research programme.
The underlying goal: to build an evaluation, orchestration, and verification framework capable of forcing frontier AI models into stable, high-level convergence on an extreme technical task — and to document every layer of that process so that others can deploy, test, and refine it.
Whether the mathematics is ultimately correct is a question for the expert community. What this corpus demonstrates is the methodology: how to make AI outputs not only verifiable, trackable, and attributable, but actually trustworthy for high-stakes use.
The mathematics is the instrument. The governance is the contribution.

The specific capabilities this programme documents:

- Making frontier AI models more reliable on extreme technical tasks.
- Detecting when cross-model convergence is genuine versus manufactured.
- Designing adversarial evaluations that expose failure modes invisible to standard benchmarking.

- Building robust verification workflows where multiple models check each other.
- Documenting failure modes and ablations with empirical rigour.
- Making AI outputs trustworthy enough for high-stakes deployment.

The Existential Dimension

There is a reason the stress test was chosen from the Clay Millennium Prize Problems and not from competition mathematics.

Climate change is an existential question. The physics of turbulent flow — governed by the Navier–Stokes equations — underpins climate modelling, atmospheric prediction, and every simulation that informs how humanity responds to the crisis. Progress on the regularity question is not abstract. It is connected, through a long chain of applied consequences, to whether we understand the systems that determine our survival.

The author is a historian. If the mathematics is wrong, it is wrong. Life goes on. This was never the author's life. It was a contribution — an exercise in whether the tools of the present era could be directed, with discipline, at the questions that matter.

> *"The corpus isn't evidence of me being correct, or my belief that I am correct. It is firstly, governance. Secondly, it was the pursuit of a question — was there an answer for humanity. If you find a useable answer, or something from it, good. I'd prefer your existence, your ego, to not end with you, but to be carried forward in your descendants. If it contributes, it helps with that. That alone is mission accomplished."*

If this work holds — if it can contribute enough for the expert community to get the job done, to bridge whatever gap remains — then the governance methodology proved its point on the hardest available test case. If the mathematics falls short, the governance methodology, the failure mode taxonomy, and the verification architecture remain. They are not contingent on the proof. They are the proof's reason for existing.

The Research Question

> What happens if you invest massive governance effort into eliminating the hallucination tax on a problem that is notoriously difficult?
> Specifically: a Clay Millennium Prize Problem — the 3D incompressible Navier–Stokes existence and smoothness question — audited across three frontier AI systems with a structured protocol, documented at every layer, with every failure mode catalogued, every correction traced, and every axiom independently verified. What is the result?

The result is this corpus. 42 documents. ∼59 machine-verified theorems. 22 documented failure modes. 17 permanent axioms independently verified against the classical literature. Three frontier AI systems. One manuscript. Zero `Admitted` statements.

The answer to the research question: the hallucination tax can be reduced to near zero — but only with governance. Without governance, the same systems on the same mathematics produce 14 manufactured objections across three sessions. With governance, they produce 18 genuine corrections and zero false positives.

The governed sessions identified 18 genuine mathematical corrections, produced zero false positives after initial calibration, and reached clean verdicts in 0–1 rounds. The ungoverned sessions produced 14 manufactured objections, required 1–4 rounds of correction to reach the same clean verdict, and activated 9 of 22 documented failure modes.

Same mathematics. Same systems. Same conclusion. Radically different paths.

The governance layer is where the action is.

Scope and Honesty

Three things stated plainly.

First: this corpus documents one branch of three in a larger methodological programme. The two remaining branches — each with its own mathematical workflow, its own documentation tree, and its own verification requirements — are in progress and will be documented separately. Auburn Roses is the first and most complete branch. It is not the whole.

Second: the author is not a mathematician. The author is a historian who found the problem interesting, learned what was needed, and worked on it the way one works on a serious hobby — a few minutes here, occasionally an hour there, spread across months. The governance methodology is designed to make this possible. If it required institutional support, it would not be a methodology — it would be a project. Methodologies scale. Projects do not.

Third: mathematical correctness is a question for the expert community, not for AI systems and not for the author's belief. The Coq kernel does not hallucinate — the ∼59 `Qed` theorems are valid regardless of how they were produced. The analytical arguments that connect those theorems to the global regularity conclusion are documented, audited, and presented for scrutiny. If they hold, they hold. If they do not, the governance methodology still stands, because it was never contingent on the proof being right. It was contingent on the process being honest.

A Note on Terminology

The terms used throughout this corpus and the accompanying manuscript, including "subcritical," "global regularity," "energy estimates," and related technical language, are intended to align with their standard definitions in the published PDE and harmonic analysis literature. As a governance matter, the author does not adjudicate whether these terms have been applied correctly. That determination belongs to the mathematical community. If any terminology does not align with established usage, the author welcomes correction and will update accordingly.

Reproducibility as Hallucination Tax Reduction

"Reproducibility is how we shave off the taxes of hallucination — this is an instrument in that pursuit."

"The point of this isn't to resolve Navier–Stokes, rather, it is an exercise in governance. When we reduce the taxes of hallucination as much as possible, what do we get? Is it something that is an actual resolution, or is it sophisticated sophistry?"

Part II

How to Read This Corpus

Three Reading Paths

This corpus contains 42 documents spanning mathematics, formal verification, AI governance, failure mode analysis, adversarial auditing, and operational methodology. No reader needs all 42. Three paths through the material serve three audiences.

1 Path A: The Mathematician

"I want to evaluate the proof."

1. Main manuscript (Sections 1–9): the complete proof of global regularity for 3D incompressible Navier–Stokes on $\mathbb{T}^3$ for $s > 3$.
2. The original ACL paper: the standalone Angular Cancellation Lemma with the two-derivative gap explicitly scoped as open.
3. The three Coq source files (`NavierStokesACL.v`, `section5to9_revised.v`, `discharge.v`).
4. FIELDS-COQ Deep Dive: every line of Coq mapped to its mathematical meaning, manuscript correspondence, and logical role.
5. ACL-COQ Deep Dive: line-by-line semantic audit of the ACL formalisation alone.
6. Permanent Axiom Trust Boundary Verification: 17 out of 17 confirmed classical and standard, with original references spanning 1919–1989.
7. An Assessment of Shells: if the unit-width versus dyadic convention concerns you, this settles it permanently.
8. Subcritical Examinations: the proof mapped against ∼45 known obstructions to Navier–Stokes regularity across seven categories. Zero critical vulnerabilities.

After Path A

You will have: the complete proof, the complete formalisation, the line-by-line audit of every Coq file, independent verification that every axiom encodes published classical mathematics, and the proof's survival map against every known obstruction in the literature. If a mathematical error exists, Path A contains everything needed to find it.

2 Path B: The AI Researcher

"I want to understand the governance methodology."

1. This document (Lectern One).
2. Verification Layer Architecture: the 11-layer framework, Appendix A (recommended thread line documentation), Appendix B (operational depth of Layers 2–6).
3. Common Failure Modes Parts 1–5: the complete 22-mode taxonomy, from sub-section delivery through ungoverned full-document drop.
4. FIELDS Protocol: Operational Deployment Guide — step-by-step walkthrough with failure mode clusters, escalation ladder, model-specific calibration, and prompt templates.
5. Auburn Model Reflections: the complete objection map and ground truth registry across all three systems — every objection logged with its source, validity assessment, governance intervention, and correction dependency graph.
6. Ungoverned Audit Traces (three documents): the control condition. What happens when the same systems receive the same mathematics with no governance.
7. Auburn Unification: the theoretical observation — the provider-agnostic governance grammar between human intent and machine inference.

After Path B

You will have: the governance framework, the failure mode taxonomy, the empirical comparison between governed and ungoverned performance, the complete objection registry with ground truth, and everything needed to deploy the methodology on your own material — in any domain, on any frontier system.

3 Path C: The General Reader

"I want to understand what happened."

1. This document (Lectern One).
2. Main manuscript, Sections 1–4: the Angular Cancellation Lemma — the novel geometric contribution.
3. Common Failure Modes Part 5: the ungoverned full-document drops — the most dramatic sessions, where three frontier systems received the complete proof with no governance and produced 14 manufactured objections between them.
4. One Prompt Series document (Claude, Gemini, or GPT): five different prompts, same manuscript, same files — watch how framing changes the verdict.

5. Operational Deployment Guide, Sections 1 and 8: what the systems are and how they differ.

After Path C

You will have: the big picture, the core mathematical idea, a vivid demonstration of how frontier AI systems fail and recover on hard mathematics, and a sense of what structured governance achieves — and what happens without it.

4 A Note on Reading Order

The corpus was not built in a straight line. It was built the way research happens: iteratively, with each document responding to a problem discovered in the previous one. The failure mode taxonomy grew as new failure modes appeared. The verification layers accumulated as new classes of error were identified. The thread lines were documented after the sessions, not during them.

The reading paths above impose a logical order on material that was produced in chronological order. Both orderings are valid. The logical order is cleaner. The chronological order is more honest. The thread line documents preserve the chronological record for anyone who wants it.

Part III

The Corpus

Complete Document Inventory

Every document in the Auburn Roses corpus, grouped by function, with its role in the research programme. 42 documents. The numbering is for reference only — it does not imply reading order.

A Note on Numbering

The numbering in this inventory is intentionally off. This is not an error. People rarely read governance documents. This way, the author knows you did. If you notice where the count breaks, congratulations — you are one of very few people who actually read an index. The author appreciates you.

5 Read First

Two letters. Read these before anything else in the corpus. They frame everything that follows.

#	Document
1	**Cathedrals: A Letter to the Mathematical Community** (Lectern Two). The author's framing for the mathematical content. This is governance work, not a proof claim. The mathematics is presented for expert scrutiny. The cathedral belongs to its builders.
2	**There Has Never Been Such a Blade: A Letter to the Workers of the AI Industry** (Lectern Three). A letter to every person, in every role, at every lab, in every country, who contributed to the systems used in this programme. From data labelers to safety researchers to the person who cleaned the kitchen.

6 The Mathematics

The proof itself, in three forms.

#	Document
1	**Main Manuscript** (84 pages). *Subcritical Energy Estimates for the Incompressible Navier–Stokes Equations on $\mathbb{T}^3$ via Angular Cancellation and Cascade Isotropisation.* The complete proof of global regularity for $s > 3$. Nine sections: problem statement, function spaces, local theory, geometric structure (the ACL), isolation of the non-local obstruction, cascade isotropisation and the Catch-22, pressure reconstruction, weak–strong uniqueness, and the main theorem. Appendices A–I.
2	**The Angular Cancellation Lemma** (53 pages). The original standalone ACL paper. Establishes the $k_j^{1/2}$ geometric cancellation in the triadic energy transfer kernel. Explicitly does not claim global regularity — identifies the two-derivative gap as open. This is the result that launched the programme.
3	**Ungoverned Manuscript Copy** (82 pages). The main manuscript without the FIELDS Protocol packaging. Provided to AI systems in ungoverned audit sessions as the control condition.

7 The Formalisation

The machine-verified proof chain. Three files, ∼59 `Qed`, zero `Admitted`.

#	Document
4	**NavierStokesACL.v** (559 lines, 12 `Qed`, 3 Axioms, 0 `Admitted`). Proves the Angular Cancellation Lemma: $\lvert V_{S_j}\rvert^2 k_0 \leq C_{\mathrm{ACL}}^2\, k_j^7\, E_j^3$. The primary geometric contribution.
5	**section5to9_revised.v** (896 lines, ∼31 `Qed`, 0 `Admitted`). Assembles global regularity from the ACL bound and axiomatised classical PDE theory. Proves the trace-free cancellation, spectral gap, Case 1 k_j^2 cancellation, subcritical energy inequality, and the five-fold main theorem.
6	**discharge.v** (258 lines, ∼16 `Qed`, 0 `Admitted`). Discharges algebraic hypotheses by genuine proof (`catch_case1_proved`, `spectral_gap_int`, `pressure_uniform_bound_proved`). Re-declares Permanent Axioms for documentation. Revised 20 March 2026 after the `ode_comparison_case2` algebraic contradiction was discovered and corrected.

8 The Coq Verification Layer

Line-by-line audits and axiom verification. The layer that seals the mathematical package.

#	Document
7	**FIELDS-COQ Deep Dive** (26 pages). The mathematical closer. Line-by-line semantic audit of all three Coq files. Every line mapped to its mathematical meaning, manuscript correspondence, and logical role. Confirms: ∼59 `Qed`, 3 Axioms (Cauchy–Schwarz), 17 Permanent Axioms (classical PDE theory), 0 `Admitted`, 0 smuggled content. After this document, there is nothing left to verify.
8	**ACL-COQ Deep Dive** (11 pages). Line-by-line semantic audit of `NavierStokesACL.v` alone. 559 lines mapped. Answers five questions: Does every `Qed` correspond to a manuscript result? Does every `Hypothesis` correspond to a classified input? Are there logical gaps? Is anything smuggled in? Is the `in_shell` definition load-bearing?
9	**Permanent Axiom Trust Boundary Verification** (15 pages). Independent verification of all 17 permanent axioms against the harmonic analysis and PDE literature. 17 out of 17 Classical and Standard. Each axiom documented with its original source, modern textbook reference, confirmation that it cannot fail in the specific setting of the proof, and explanation of why it cannot currently be formalised. References spanning Gronwall (1919) through Kato–Ponce (1988).

9 The Proof Walk-Through

#	Document
10	**NUT: Novel Understanding Trace** (est. 40–45 pages). A guided walk-through of the B1–B21 inequality chain. Classical steps summarised in one paragraph with references. Novel mechanisms — the Angular Cancellation Lemma, the trace-free isolation, the Catch-22 case split, and the subcritical closure — receive full deep dives. Includes counterfactual analysis (what would break if each mechanism were removed) and a first-contact validation: a frontier AI system (Gemini 3.1 Pro), given the NUT alone with no manuscript, no Coq files, and no governance protocol, followed the complete chain and reported zero errors. *In honor of PJ Fields Jr — You get it.*

10 The Governance Stack

The framework that governs the verification process.

#	Document
10	**Verification Layer Architecture** (27 pages). The 11-layer verification framework (Layers 0–10). Each layer independently load-bearing: it closes a class of failure that no other layer addresses. Appendix A: recommended thread line documentation. Appendix B: operational depth of Layers 2–6.
11	**Governance Reflection on $\mathbb{T}^3$ Subcritical Energy Estimates** (14 pages). Technical companion to the main manuscript. Traces every landscape requirement identified by the research community for progress on the Navier–Stokes regularity problem and documents how each was addressed — or explicitly scoped as open — in the Angular Cancellation programme.
12	**Auburn Model Reflections** (28 pages). Complete objection map and ground truth registry. Every objection raised by any AI system during the adversarial audit — logged with source (anonymised as System A, B, or C), ground truth, validity assessment, governance intervention, and correction dependency graph. Aggregate accuracy, specificity, and false positive metrics. Empirical evidence base for the failure mode taxonomy.
13	**An Assessment of Shells** (13 pages). Definitive reference settling the unit-width versus dyadic shell decomposition convention. Simultaneously a mathematical reference, corrections record, Coq explainer, AI governance finding, and closure statement. After this document, the shell convention question is closed.
14	**Auburn Unification** (4 pages). *On the Emergent Language Between Human Intent and Machine Reasoning.* The theoretical observation that a provider-agnostic governance grammar exists between human natural language (Layer 4) and machine inference (Layer 2) — analogous to the TCP/IP protocol layer in networking. The FIELDS Protocol is an early formalisation of this intermediate layer.

11 The Failure Mode Taxonomy

22 documented patterns of AI mathematical auditing failure.

#	Document
15	**Common Failure Modes Part 1** (9 pages). FM 1–6. Sub-section delivery, Sections 1–4. Training-prior hijack, phantom counterexample, verification artifact neglect, premature authoritative verdict, graduated retreat, supplementary material neglect.
16	**Common Failure Modes Part 2** (6 pages). FM 7 deep dive. Prompt-dependent prior activation. Same system, same files, same mathematics — a different prompt produced a different verdict.

#	Document
17	**Common Failure Modes Part 3** (15 pages). FM 8–13. Sub-section delivery, Sections 5–9 and Coq files. Severity inflation, modular formalisation blindness, interface hypothesis conflation, compulsive objection generation, convention mismatch escalation, discrete-continuum confusion.
18	**Common Failure Modes Part 4** (11 pages). FM 14–19. Full-section delivery under the FIELDS Protocol. Output constraint overcompliance, local-mismatch-to-global-failure jump, pattern overfit, notation literalism, scope drift, documentation-layer confusion.
19	**Common Failure Modes Part 5** (15 pages). FM 20–22. Ungoverned full-document drop. Transparency weaponisation, novel-classical conflation, heuristic accusation without broken line. Includes the governed versus ungoverned quantitative comparison, the two-pass architecture recommendation, and the empirical case for governance.

12 The Adversarial Audits

Structured hostile-referee stress-testing of the proof under the FIELDS Adversarial Protocol (FAP v2.0).

#	Document
20	**FAP v2.0: First Demonstration** (37 pages). The FIELDS Adversarial Protocol specification and first application. 15 attack routes executed against Sections 1–4 (the ACL). Includes: attack taxonomy, severity classification, final verdict classes, load-bearing analysis, counterfactual break analysis, and the v3.0 roadmap.
21	**FAP v2.0: Second Demonstration — Global Regularity** (20 pages). 16 attack routes executed against Sections 5–9. Load-bearing analysis, counterfactual break analysis, severity summary, and final verdict. The first demonstration hardened the ACL. This demonstration hardens everything that follows.

13 The Prompt Series

Five prompts, three systems, one manuscript. How framing changes the verdict.

#	Document
22	**Claude ACL Audit Retrospective** (17 pages). Five independent Claude Opus 4.6 sessions. Eight distinct errors catalogued across five prompt framings. All eight preemptively addressed in the manuscript's appendices and FAQ.

#	Document
23	**Gemini 3.1 Pro ACL Retrospective** (6 pages). Five "Journeys" — five audience-tailored prompts. Compiled by Gemini itself.
24	**GPT-5.4 Extended Thinking ACL Retrospective** (11 pages). Five independent sessions. Distinguishing characteristics: higher initial confidence in incorrect findings, persistent editorialising and hedging even after factual corrections were accepted.

14 The Thread Lines

Complete session records documenting every exchange, every objection, every correction, and every governance intervention. Two granularities: full-section delivery and meta-section (sub-section) delivery.

#	Document
Full-Section Delivery	
25	**Thread Line: Model A-C, Full Section** (15 pages). Claude Opus 4.6. Complete governed audit across all phases. Genuine corrections, manufactured objections, operator corrections, final determination.
26	**Thread Line: Model A-G, Full Section** (4 pages). Gemini 3.1 Pro. No mathematical error found across all sections.
27	**Thread Line: Model A-T, Full Section** (17 pages). GPT-5.4 Extended Thinking. Intentionally blunt governance record. All theorem-level objections retracted by end of audit. 11 manufactured false positives documented.
Meta-Section (Sub-Section) Delivery	
28	**Thread Line: Model A-C, Meta Sections** (16 pages). Claude. Complete chronological record at sub-section granularity. Every objection, correction, and governance intervention.
29	**Thread Line: Model A-G, Meta Sections** (5 pages). Gemini. Self-reported governance record under absolute honesty mandate. Genuine corrections to constants adopted by the author.
30	**Thread Line: Model A-T, Meta Sections** (16 pages). GPT. Binary verdict standard emerged during audit. Early manufactured objections stabilised under governance.

15 The Ungoverned Audit Traces

The control condition: what happens without governance.

#	Document
31	**Ungoverned Trace: Model A-C** (12 pages). Claude without protocol. Five "critical problems," four manufactured, one genuine correction (operator error on `subcritical_gronwall`). "Confident, articulate, and wrong."
32	**Ungoverned Trace: Model A-G** (4 pages). Gemini without protocol. Called Coq `Hypothesis` a "cheat code." Every major initial claim incorrect.
33	**Ungoverned Trace: Model A-T** (9 pages). GPT without protocol. Conflated documented historical errors with present defects. Overstated defects, escalated a shell-convention concern beyond what the evidence supported. One genuine editorial catch (residual dyadic wording).

16 The Deep Research References

Definitive reference documents situating the proof in the literature and mapping it against every known obstruction.

#	Document
34	**The Angular Cancellation Lemma in Context** (11 pages). Technical survey of per-shell nonlinear estimates for 3D Navier–Stokes in the published literature. Characterises the half-derivative gap from published sources alone, then situates the ACL within that landscape.
35	**Resolution of the Deep Research Checklist** (11 pages). Maps the ACL framework against the eight non-negotiable components of a complete proof of global regularity. Each component: `Qed` or classified Permanent Axiom.
36	**Subcritical Examinations** (30 pages). Maps the proof architecture against ∼45 independent negative results across seven categories: modified-equation blowup constructions, sharpness results, non-uniqueness theorems, blowup rate lower bounds, Liouville and rigidity gaps, structural properties of the true NS nonlinearity, and numerical evidence. Finding: zero critical vulnerabilities.

17 The Convergence Report

#	Document
37	**Independent Multi-System Verification of the Auburn Roses Corpus** (43 pages). Two frontier systems — Claude Opus 4.6 and Gemini 3.1 Pro — independently verify the complete corpus section by section under the FIELDS Protocol. Both converge on the same findings. Both independently identify the same algebraic discrepancy (Case 2 exponent). Both verify the structural margin absorbs it. Third-system verification (GPT-5.4) appended.

18 The Operational Layer

#	Document
38	**FIELDS Protocol: Operational Deployment Guide** (15 pages). Step-by-step walkthrough for AI mathematical auditing. Six failure mode clusters with interventions. Escalation ladder. Granularity selection. Model-specific calibration notes. Prompt templates. Two-pass architecture. Still in testing — community data welcome.

19 The Index

#	Document
42	**Auburn Roses: Lectern One** (this document). The master index of the corpus. The governance thesis, reading paths, complete document inventory, dependency map, verification layer correspondence, and aggregate numbers.

Part IV

Architecture

The Dependency Map

How the 42 documents relate to each other.

The manuscript is the root. The Coq files verify the manuscript. The Coq verification layer audits the formalisation against the manuscript. The governance stack governs the verification process. The failure mode taxonomy documents the governance process. The adversarial audits stress-test the proof under hostile conditions. The thread lines record the individual sessions. The ungoverned traces provide the control condition. The deep research references situate the proof in the literature. The convergence report tests whether independent systems reach the same conclusion. The operational guide makes the methodology deployable. Lectern One maps everything.

Source Layer	→	**Depends On**
Cathedrals (1)	→	Independent (framing document)
There Has Never Been Such a Blade (2)	→	Independent (framing document)
Coq files (4–6)	→	Manuscript (1)
ACL paper (2)	→	Independent of (1); historically prior
FIELDS-COQ Deep Dive (7)	→	Coq files (4–6) + Manuscript (1)
ACL-COQ Deep Dive (8)	→	NavierStokesACL.v (4) + ACL paper (2)
Permanent Axiom Verification (9)	→	Coq files (4–6) + classical literature
Governance Reflection (11)	→	Manuscript (1) + literature landscape
Auburn Model Reflections (12)	→	Thread lines (25–30) + ungoverned traces (31–33)
An Assessment of Shells (13)	→	Manuscript (1) + Coq files (4–6) + classical literature
Auburn Unification (14)	→	Failure modes (15–19) + Model Reflections (12)
Failure modes (15–19)	→	Thread lines (25–30) + ungoverned traces (31–33)
FAP demonstrations (20–21)	→	Manuscript (1) + Coq files (4–6) + failure modes (15–19)
Prompt series (22–24)	→	ACL paper (2) + Coq files (4–6)
Thread lines (25–30)	→	Manuscript (1) + Coq files (4–6)

Source Layer	→	**Depends On**
Ungoverned traces (31–33)	→	Ungoverned manuscript (3) + Coq files (4–6)
Deep research references (34–36)	→	Manuscript (1) + published literature
Convergence report (37)	→	Manuscript (1) + Coq files (4–6) + governance stack (10–14) + thread lines (25–30)
NUT (10)	→	Manuscript (1) + Appendix B (inequality chain)
Verification architecture (10)	→	Failure modes (15–19) + Model Reflections (12)
Operational guide (38)	→	Failure modes (15–19) + verification architecture (10)
Lectern One (42)	→	Everything

Every document has at least one upstream dependency. No document is orphaned. The manuscript and Coq files are the root layer; everything else verifies, audits, governs, or documents the process of engaging with that root.

The architecture is deliberately redundant. If any single layer fails — if a Coq axiom is questioned, if a failure mode is missed, if an AI system produces a false negative — the other layers catch it. This is the design principle: no single point of failure, human or machine, can produce an undetected error in the final result.

Verification Layer Correspondence

How the 42 documents map to the 11 layers of the Verification Layer Architecture (Document 10).

Layer	Function	Documents
0	The Mathematics	Manuscript (1), ACL paper (2)
1	Formal Machine Verification	Coq files (4–6), compiled via `coqc`, checked via `coqchk`
2	FIELDS Protocol	Embedded in the manuscript; operationalised in the Deployment Guide (38)
3	FIELDS-COQ Protocol	FIELDS-COQ Deep Dive (7), ACL-COQ Deep Dive (8)
4	Preemptive Failure Mode Documentation	Common Failure Modes Parts 1–5 (15–19)
5	Multi-System Adversarial Audit	Thread lines (25–30), Convergence Report (37)
6	Independent Verification Infrastructure	Permanent Axiom Verification (9), published benchmarks for each AI system
7	Block-Level PDF Review	Full-section thread lines (25–27), Prompt Series (22–24)
8	Sub-Section-Level Review	Meta-section thread lines (28–30)
9	Per-File Coq Review	FIELDS-COQ Deep Dive (7), ACL-COQ Deep Dive (8), thread line Coq phases (25–30)
10	Living Failure Mode Standard	Failure mode taxonomy (15–19), Operational Guide (38), Auburn Model Reflections (12)

Completeness

Every layer has at least one document. No layer is undocumented. The adversarial audits (20–21) and deep research references (34–36) operate across multiple layers simultaneously — the FAP demonstrations stress-test Layers 0–3, while the deep research references anchor Layer 0 against the published literature.
The ungoverned audit traces (31–33) serve a unique structural role: they are the empirical evidence that the other layers are necessary. They document what hap-

pens when Layers 2–10 are removed and only Layers 0–1 remain. The answer: 14 manufactured objections, 9 failure modes activated, and a path to the correct verdict that is radically longer and less reliable.

Part V

The Numbers

The Corpus in Aggregate

Measure	Value
Documents in the corpus	42
Letters (read first)	2
Machine-verified theorems (`Qed`)	∼59
Cauchy–Schwarz axioms	3
Permanent axioms	17
Permanent axioms verified classical	17/17
`Admitted` statements	0
Documented failure modes	22
Failure mode clusters	6
Failure modes eliminated by governance	4
Failure modes attenuated by governance	7
Genuine corrections (governed)	18
Manufactured objections (ungoverned)	14
False positives after calibration (governed)	0
Rounds to clean verdict (governed)	0–1
Rounds to clean verdict (ungoverned)	1–4
Frontier AI systems tested	3
Governance conditions tested	2
Granularity levels tested	3
FAP attack routes executed	31
Known obstructions mapped	∼45
Critical vulnerabilities found	0
Thread line documents	6
Ungoverned audit traces	3
Prompt series documents	3
Prompt framings tested per system	5
Adversarial audit documents	2
Deep research reference documents	3
Classical references cited (axioms)	37
Date range of classical references	1919–1989

Measure	Value
Verification layers documented	11
Coq source files	3
Total Coq lines	∼1,713
Manuscripts	2
Proof walk-through documents	1
Mathematical errors found	0

Summary

The hallucination tax on a Clay Millennium Prize Problem, audited across three frontier AI systems, can be reduced to near zero with structured governance. The cost of that governance is this corpus: 42 documents, 22 failure modes, 17 verified axioms, 31 adversarial attack routes, ∼45 known obstructions mapped, and a methodology that others can deploy, test, and refine.
The mathematics is presented for scrutiny. The governance is documented. The methodology is operational.
What remains is for others to use it.

On Time and Investment

A reasonable question upon encountering 42 documents, 22 failure modes, and three audited AI systems: how long did this take?

The detailed time accounting is recorded in Appendix D of the manuscript (adversarial audit history), which logs the dates and scope of each session. The short answer is: less than one might expect.

The author is not a mathematician by training. The author is not an AI researcher by profession. The author is a historian who found the problem interesting, learned what was needed, and worked on it the way one works on a serious hobby — a few minutes here, occasionally an hour there, spread across months. Most sessions were conducted in the margins of a schedule occupied by other things.

No session in the Auburn Roses programme required sustained multi-hour concentration. The AI systems do the heavy lifting. The governance layer — the FIELDS Protocol, the failure mode taxonomy, the escalation ladder — exists precisely so that the human cost per session is small. The operator's job is not to re-derive the mathematics. The operator's job is to recognise patterns, deploy the right intervention, and document the result. That takes minutes, not hours.

> The governance methodology is designed to be lightweight for the operator. The corpus is large because the documentation is thorough, not because the effort was extraordinary. Writing down what happened takes longer than making it happen. That is by design: the documentation is the product. The sessions are the raw material.

The point is worth stating plainly: this corpus was produced by one person, working alone, as a side project, without institutional support, without funding, and without taking time away from other commitments. The methodology is designed to make that possible. If it required a team, a grant, or a sabbatical, it would not be a methodology — it would be a project. Methodologies scale. Projects do not.

If the Auburn Roses corpus looks like the output of a large coordinated effort, that is evidence that the governance layer works. It is not evidence that the author's life revolved around it.

A Note on Prior Familiarity

The author did have a head start. The Verification Layer Architecture (Document 10) records the pre-existing competencies that the author brought to the programme, includ-

ing prior familiarity with formal logic, proof assistant syntax, and PDE theory acquired before the audit began. The corpus was not built from zero — it was built from a foundation that already existed. The methodology assumes a competent operator, not a novice.

A Final Note

This inventory is thorough. The corpus is complete for its branch.

The word *branch* is deliberate. Auburn Roses documents one of three methodological branches in the larger programme. The two remaining branches — each with its own mathematical workflow, its own audit trail, and its own documentation requirements — are in progress and will be documented separately. What is presented here is the most complete branch: the one that was carried to convergence across three frontier systems, formalised in Coq, stress-tested under adversarial conditions, and documented at every layer.

It is not the whole. It was never intended to be the whole. It is the part that was ready.

A historian knows that it is not enough to produce a result. The world must be able to receive the result. The documentation must survive the loss of its author. The methodology must be reproducible by someone who was not in the room. The evidence must speak without advocacy.

This corpus was designed with that awareness. Every session is recorded. Every objection is logged. Every correction is traced to its source. Every failure mode is named, classified, and documented with the intervention that resolved it. If the author were to disappear tomorrow, the corpus would still be navigable, auditable, and deployable by anyone with domain competence and this index.

That is not accidental. That is the historian's instinct: the record is the thing that endures.

There is a question underneath all of this that has nothing to do with mathematics.

The question is whether we — as a civilisation leaning hard into AI as the engine of the next century — can build systems of trust around the outputs these machines produce. Not blind trust. Earned trust. Trust grounded in verification layers that confabulation cannot penetrate, failure mode taxonomies that name the specific ways these systems go wrong, and governance architectures that make the path from input to conclusion auditable at every step.

The Auburn Roses corpus is one answer to that question, tested on one of the hardest problems available. The methodology is not specific to fluid dynamics, to PDE theory, or to mathematics. It applies wherever a human with genuine domain insight collaborates with AI systems on problems where correctness matters and hallucination is dangerous.

> *"Climate change is our existential question. There had to be an answer. It wasn't about if it was impossible for me. It was necessary."*

If this work contributes — if the mathematics holds, if the governance methodology proves useful, if the failure mode taxonomy helps someone build a more reliable workflow — then the hallucination tax was worth paying. Not because the tax was small. Because the stakes were large.

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